HDOS RWANDA™

HUMAN DEVELOPMENT INTELLIGENCE OPERATING SYSTEM

TRANSFORMING NATIONAL DATA AND HUMAN GOALS INTO EVIDENCE-BASED LIFE OPPORTUNITIES

SOLUTION SUMMARY

HDOS Rwanda connects individuals' goals and personal profiles with trusted national statistics, education and labour-market intelligence, and verified opportunities to generate explainable development pathways, support applications, and track progress.

COMPLETE INNOVATION, DATA INTELLIGENCE, APPLICATION, SERVICE & SYSTEM ENGINEERING BOOK

NISR 2026 HACKATHON

TRACK 3: OPEN INNOVATION

Education • Employment • Youth Development • Poverty Reduction • Digital Transformation • Human Capital Development

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1. EXECUTIVE SUMMARY

Rwanda has made important progress in education, digital transformation, innovation, employment promotion and economic development. Nevertheless, students, graduates, job seekers, entrepreneurs and professionals still experience difficulties making informed decisions about their education, careers, skills, business activities and economic future.

People continuously ask:

- Which career should I pursue?
- Which university or TVET institution should I attend?
- Which academic programme is appropriate for my abilities and ambitions?
- Which skills should I develop?
- Which jobs are available and suitable for me?
- Which scholarships and educational opportunities can I access?
- Which business opportunities have potential?
- How can I improve my income?
- How can I achieve my long-term goals?

Information exists across government institutions, education providers, employment platforms, research publications, online services and artificial intelligence systems. However, people still need to collect, compare, interpret and convert this information into practical decisions.

HDOS Rwanda™ introduces an integrated approach to this challenge.

HDOS stands for **Human Development Intelligence Operating System**. It is designed to connect personal goals with national statistics, education intelligence, labour-market information, economic context, opportunity information and artificial intelligence.

It does not simply provide information. It helps users understand their present position, identify development gaps, compare available pathways, discover opportunities, prepare applications, take action and monitor progress.

The HDOS development cycle is:

GOALS → PERSONAL PROFILE → DATA → INTELLIGENCE → DECISIONS → OPPORTUNITIES → ACTION PLANS → PROGRESS → OUTCOMES

The platform is designed for both individual users and institutions. Individual users receive personalized development guidance, while institutions may use aggregated, appropriately governed intelligence to support education planning, workforce development and human-capital initiatives.

The initial implementation focuses on proving that reliable national information and individual circumstances can be combined to create useful, explainable and measurable development guidance.

The long-term ambition is to develop HDOS into a scalable human-development intelligence platform beginning in Rwanda and potentially expanding across Africa and international markets.

2. PROJECT IDENTITY, VISION AND MISSION

2.1 Official project name

HDOS Rwanda™ — Human Development Intelligence Operating System

The name represents an integrated intelligence environment that helps people organize their goals, understand opportunities and make informed development decisions.

2.2 Vision

To build an intelligent system that enables individuals and institutions to transform human potential into measurable development outcomes through evidence-based decisions.

2.3 Mission

To connect personal aspirations with reliable national and global information, intelligent analysis, practical development pathways and accessible opportunities.

2.4 Strategic objectives

HDOS aims to:

1. Improve access to relevant education and career information.
2. Help users understand their skills, interests and development needs.
3. Connect education and training pathways with labour-market information.
4. Help individuals discover and assess relevant opportunities.
5. Support users in preparing and managing applications.
6. Provide evidence-based recommendations and personalized roadmaps.
7. Support institutions with aggregated human-capital intelligence.
8. Promote more informed entrepreneurship and economic decisions.

9. Measure user progress through practical milestones and outcomes.
10. Build an adaptable intelligence platform that can expand beyond Rwanda.

2.5 Project ownership and institutional independence

HDOS Rwanda is an independent innovation project associated with IRAVUGA TECHNOLOGIES GROUP.

Its development may use publicly available information from national institutions, but this does not imply government ownership, official endorsement, direct institutional partnership or access to restricted government databases.

Any formal partnership, data-sharing agreement, institutional deployment or government integration must be established separately.

3. RWANDA'S HUMAN DEVELOPMENT CHALLENGE

3.1 The national opportunity

Rwanda continues to invest in:

- Education and skills development.
- Youth employment.
- Technical and vocational training.
- Entrepreneurship and innovation.
- Digital transformation.
- Private-sector development.
- Human-capital development.
- Improved public service delivery.

These priorities create opportunities for individuals, educational institutions, employers and innovators.

However, access to information alone does not automatically enable people to make effective decisions.

A student may know that information technology is growing but may not know which subjects to prioritize, which institutions offer relevant programmes, which skills employers require or which opportunities are realistic.

A graduate may possess an academic qualification but lack practical skills, experience, professional networks or knowledge of available opportunities.

An entrepreneur may identify a business idea without enough information about market demand, customers, competition, financing, skills or regional conditions.

Therefore, development requires more than information access. It requires the ability to interpret information in relation to individual circumstances and practical objectives.

3.2 The decision-making gap

The core national challenge addressed by HDOS is the gap between:

Available Information and **Actionable Personal Development Decisions**.

This gap appears in several areas.

Education

Students may choose programmes without understanding their requirements, costs, progression routes or employment relevance.

Employment

Job seekers may struggle to understand labour-market demand, occupational requirements and the skills needed to improve employability.

Entrepreneurship

Potential entrepreneurs may lack accessible business intelligence and structured pathways from an idea to a viable business.

Economic development

Individuals may not know which professional, educational or entrepreneurial pathways could improve their economic position.

Opportunity access

Relevant scholarships, internships, grants, jobs and training programmes may be published by different institutions and may have different requirements and deadlines.

3.3 The required transformation

HDOS seeks to transform:

- Unorganized information into structured intelligence.
- General recommendations into personalized pathways.
- Development ambitions into measurable goals.
- Opportunity discovery into eligibility assessment.
- Application preparation into organized execution.
- Individual actions into trackable progress.

HDOS is designed to support this process, not to claim that unemployment, poverty or other structural challenges can be solved by one software platform.

4. PROBLEM STATEMENT AND EXISTING DIGITAL LANDSCAPE

4.1 Existing digital services

Different platforms already solve important parts of the human-development challenge.

Platform category	Main value	Remaining integration opportunity	
Search engines	Information discovery	Connect information with user goals and requirements	
Social media	Communication and public discussion	Improve information quality and traceability	
Learning platforms	Courses and learning resources	Connect learning with occupations and real opportunities	
Job platforms	Vacancy discovery and matching	Connect jobs to broader education and development pathways	
Scholarship portals	Study funding announcements	Connect scholarship requirements with profiles and applications	
AI assistants	Explanation and conversational assistance	Connect answers with verified data, workflows and progress	
Statistical portals	National and sector indicators	Translate aggregated statistics into individual decision support	

It would be inaccurate to claim that no platform provides career matching, scholarship discovery or labour-market information. Rwanda already has services such as Kora, the Rwanda Labour Market Information System and the Higher Education Council's scholarship announcements.

The innovation opportunity for HDOS is to integrate these different information domains around an individual's long-term development goals.

4.2 The central problem

Existing information services may operate independently, while users must make the connections themselves.

A student may need to:

1. Visit an education information source.
2. Search for an institution.
3. Identify a suitable academic programme.
4. Search for its admission requirements.
5. Compare the programme with career requirements.
6. Find relevant scholarships.
7. Prepare documents.
8. Search for internships.
9. Identify skill gaps.
10. Build a realistic development plan.

HDOS proposes to coordinate these steps through a connected intelligence workflow.

4.3 Problem statement

Individuals and institutions need a unified, evidence-based decision-support environment that connects personal goals with education, employment, skills, economic information and development opportunities, then converts these connections into practical and trackable action plans.

5. HDOS RWANDA: THE INTEGRATED SOLUTION

HDOS Rwanda is a human-development intelligence platform that combines personal information, national statistics, education data, workforce information, economic intelligence, opportunity records and AI-supported analysis.

Its principal purpose is to guide decisions and help users take practical actions.

5.1 Main operating workflow

1. **Goal definition:** The user identifies what they want to achieve.
2. **Personal assessment:** The system develops a profile of education, skills, interests, experience and constraints.
3. **Data connection:** Relevant national, educational, occupational and opportunity records are retrieved.
4. **Intelligence generation:** HDOS identifies relevant pathways, opportunities and development gaps.
5. **Decision comparison:** Alternatives are compared using transparent criteria.
6. **Action planning:** The system creates a development roadmap.
7. **Opportunity execution:** The user discovers, assesses and prepares applications.
8. **Progress monitoring:** Milestones, activities and application outcomes are recorded.
9. **Continuous improvement:** Recommendations are updated when the user's profile or external information changes.

5.2 Core transformation

GOAL → UNDERSTAND → COMPARE → CHOOSE → PREPARE → ACT → MEASURE → IMPROVE

This sequence is the central operating identity of HDOS.

6. CORE INNOVATION AND OPERATING PHILOSOPHY

6.1 From answering questions to guiding decisions

Traditional information systems commonly follow:

QUESTION → INFORMATION

HDOS follows:

GOAL → ANALYSIS → EVIDENCE → DECISION OPTIONS → PERSONALIZED ROADMAP → ACTION → PROGRESS TRACKING

The system is intended to help individuals understand not only what options exist, but also why an option may suit their circumstances and what they should do next.

6.2 The integrated intelligence model

HDOS combines six dimensions:

1. Personal intelligence.
2. Education intelligence.
3. Workforce intelligence.
4. Opportunity intelligence.
5. Economic intelligence.
6. Future intelligence.

These dimensions work together instead of operating as unrelated services.

6.3 Explainability as a core design rule

A recommendation should explain:

- Which user information was considered.
- Which external data sources were used.
- When the data was published or retrieved.
- Which eligibility requirements were satisfied.
- Which requirements were not satisfied.
- Which assumptions were made.
- What information is missing.
- Why the recommendation was generated.
- What the user can do to improve the result.

HDOS must avoid presenting unsupported predictions as facts.

7. TARGET USERS AND BENEFICIARIES

7.1 Students

Students can use HDOS to explore academic programmes, career pathways, skills and scholarship opportunities.

Potential benefits:

- Better-informed education choices.
- Understanding of academic and professional requirements.
- Personalized learning and career roadmaps.
- Scholarship discovery and application preparation.

7.2 Youth

Young people can use HDOS to explore training, skills development, internships, employment and entrepreneurship.

Potential benefits:

- Improved access to opportunities.
- Structured personal development plans.
- Better understanding of job requirements.
- Career-readiness guidance.

7.3 Graduates

Graduates can use HDOS to evaluate their employability and professional development requirements.

Potential benefits:

- Skills-gap assessment.
- Internship and job discovery.
- CV improvement.
- Interview preparation.
- Career-transition assistance.

7.4 Job seekers and professionals

They can compare professional pathways, identify additional qualifications, investigate labour-market trends and plan career development.

7.5 Entrepreneurs and SMEs

They can explore sectors, business opportunities, market information, skills requirements and business-development pathways.

7.6 Educational institutions

Schools, universities and TVET institutions can use institutional services for student guidance, programme intelligence, aggregated skills analysis and career-support activities.

7.7 Employers and enterprises

Employers can use appropriate HDOS services to understand occupational skills, identify development needs and support workforce planning.

7.8 Government and development partners

With appropriate permissions, data-sharing agreements, privacy controls and institutional contracts, HDOS may provide aggregated intelligence to support workforce planning, policy research and human-capital development.

It must not expose private individual profiles through public or institutional dashboards without a lawful basis and appropriate authorization.

8. THE SIX ORIGINAL CORE INTELLIGENCE MODULES

The six original modules remain the foundation of HDOS.

8.1 HDOS Goal Engine™

Purpose

The Goal Engine understands, organizes and structures individual goals into achievable development pathways.

Original examples

- Become a Software Engineer.
- Start an Agribusiness.
- Increase income.
- Become a Data Scientist.
- Study Abroad.

Enhanced capabilities

- Goal creation and categorization.
- Short-, medium- and long-term goals.
- Goal prioritization.
- Milestone definition.
- Dependency identification.
- Resource and time planning.
- Goal feasibility review.
- Progress tracking.
- Goal adjustment.
- Multiple-goal management.

Output

A personalized goal structure with milestones, timelines, suggested actions and progress indicators.

8.2 HDOS Life DNA™

Purpose

Life DNA builds a dynamic personal development profile.

Original profile dimensions

- Skills.
- Interests.
- Personality preferences.
- Learning style.
- Financial situation.
- Risk tolerance.

Additional profile dimensions

- Education history.
- Academic qualifications.
- Professional experience.
- Projects and portfolio.
- Languages.
- Certifications.
- Career preferences.
- Geographic preferences.
- Learning resources.
- Time availability.

- Accessibility requirements.
- Development constraints.

Output

A structured profile used to support career, education, skills and opportunity recommendations.

Privacy requirement

Users must understand why information is collected and how it will be used. Financial information, personality information and other sensitive or high-impact profile attributes should not be collected unnecessarily or used in unexplained automated decisions.

8.3 HDOS Workforce Intelligence™

Purpose

The Workforce Intelligence module uses labour-market information to help users understand occupational structures and changing skills requirements.

Original capabilities

- High-demand careers.
- Skills shortages.
- Emerging opportunities.
- Employment forecasts.

Enhanced capabilities

- Occupation catalogue.
- Industry mapping.
- Occupational qualifications.
- Required technical and transferable skills.
- Employment indicators.
- Regional workforce information.
- Sector skills analysis.
- Graduate employment information.
- Wage indicators where reliable data exists.
- Historical labour trends.
- Future skills scenarios.

Output

Occupation profiles, skill requirements, relevant pathways and evidence-based workforce insights.

Forecasts must be labelled as estimates or scenarios and must identify their source, assumptions and uncertainty.

8.4 HDOS Opportunity Intelligence™

Purpose

This module identifies education, employment, business and development opportunities.

Original categories

- Scholarships.
- Education opportunities.
- Businesses.
- Growth industries.
- Regional opportunities.

Expanded categories

- Jobs.
- Internships.
- Apprenticeships.
- Scholarships.
- Study loans.
- University admissions.
- TVET programmes.
- Professional training.
- Certifications.
- Grants.
- Fellowships.
- Innovation competitions.
- Incubation programmes.
- Entrepreneurship support.
- Research opportunities.
- Remote work opportunities.

Output

A searchable, categorized and continuously reviewed opportunity catalogue.

Every opportunity should contain its publisher, original link, publication date, deadline, eligibility rules, requirements and last verification date where available.

8.5 HDOS Decision Intelligence™

Purpose

Decision Intelligence compares available options and helps users select pathways that align with their goals and circumstances.

Original assessment factors

- Demand.
- Cost.
- Time.
- Personal compatibility.
- Future growth.
- Success probability.

Improved design

The original factors are retained, but the term "success probability" should not be used unless HDOS has a validated predictive model with sufficient outcome data.

The initial system should instead provide an explainable compatibility score, an evidence-confidence measure and explicit limitations.

Output

- Ranked options.
- Reasons for each ranking.
- Advantages and disadvantages.
- Estimated costs where reliable.
- Duration and requirements.
- Skills gaps.
- Alternative options.
- Recommended next steps.

8.6 HDOS Future Engine™

Purpose

The Future Engine analyzes technological change, future jobs, emerging industries and future skills demand.

Original capabilities

- Future jobs.
- Emerging industries.
- Technological change.

- Future skills demand.

Expanded capabilities

- AI and automation trends.
- Green-economy skills.
- Digital transformation.
- E-mobility.
- Advanced manufacturing.
- Data and software skills.
- Future agriculture technologies.
- Global workforce scenarios.
- Emerging occupational categories.

Output

Future-skills reports and development recommendations.

Future Engine outputs should distinguish historical observations, current evidence and forward-looking scenarios.

9. EXPANDED HDOS INTELLIGENCE AND SERVICE MODULES

The original six modules remain the core intelligence layer. This edition introduces additional service modules that extend their practical value.

9.1 HDOS Learn™

Introduced in the original national-pilot roadmap.

Provides:

- Learning recommendations.
- Academic roadmaps.
- Course and certification discovery.
- Learning milestone tracking.
- Skills-development progress.

9.2 HDOS Career™

Introduced in the original national-pilot roadmap.

Provides:

- Career planning.
- Career-readiness analysis.
- Job-readiness roadmaps.
- Occupational requirements.
- Professional development guidance.

9.3 HDOS Skills Gap Engine™

Compares a user's existing skills with skills required by a selected occupation, programme or opportunity.

Outputs:

- Current skill profile.
- Required skills.
- Missing skills.
- Skill importance.
- Recommended learning activities.
- Suggested projects.
- Progress measurements.

9.4 HDOS Business™

Introduced in the original national-development roadmap.

Supports entrepreneurs, SMEs and startups through:

- Business opportunity assessment.
- Sector intelligence.
- Market research.
- Competitor analysis.
- Business model guidance.
- Startup roadmaps.
- Risk analysis.
- Growth planning.

Business recommendations should not be presented as guaranteed profitability or investment advice.

9.5 HDOS Income Intelligence™

Introduced in the original national-development roadmap.

Uses appropriate income, employment and regional statistics to support:

- Income-improvement planning.
- Career-growth recommendations.
- Skills-based earning pathways.
- Professional development options.
- Business-development guidance.

The system should never guarantee an income increase.

9.6 HDOS National Intelligence Dashboard

Designed for authorized institutional and policy users.

Potential capabilities:

- Aggregated skills mapping.
- Workforce indicators.
- Education and skills trends.
- Youth employment indicators.
- Regional opportunity mapping.
- Institutional programme analysis.
- Development progress monitoring.

All outputs must use appropriate aggregation, suppression of small groups where necessary and authorization controls.

9.7 HDOS Application Portal™

Purpose

To help users move from discovering an opportunity to understanding its requirements, preparing documents, completing an application and tracking its progress.

Supported application categories:

- Jobs.
- Internships.
- Scholarships.
- Study loans.
- University admissions.
- TVET programmes.
- Professional training.

- Grants.
- Fellowships.
- Competitions.
- Incubation programmes.

Core functions

1. Applicant profile.
2. Opportunity explorer.
3. Eligibility assessment.
4. Requirement checklist.
5. Document matching.
6. Application preparation.
7. Application review.
8. User-authorized submission.
9. Submission confirmation.
10. Application status tracking.
11. Interview and response management.
12. Application history and archive.

Automation must respect the official publisher's rules. HDOS must not bypass CAPTCHA, authentication, access restrictions or legal declarations.

A user must personally review and approve statements, declarations, signatures and other commitments that only the user can truthfully make.

9.8 HDOS Secure Document Vault™

A dedicated service for organizing and reusing the user's application documents.

Core document categories:

1. CV or Résumé.
2. Diploma.
3. Academic Transcript.
4. Recommendation Letter.
5. Passport.
6. National ID.
7. Certificates.

Additional categories:

- Cover Letter.
- Motivation Letter.
- Portfolio.
- Birth Certificate.

- Proof of Address.
- Language Certificate.
- Professional Licence.
- Academic Equivalence.
- Other Supporting Documents.

Capabilities

- Upload and secure storage.
- Document categorization.
- Search and filtering.
- Preview and download.
- File replacement.
- Version management.
- Expiry tracking where relevant.
- Opportunity-specific document matching.
- Missing-document alerts.
- Application-package creation.
- Access history.

Uploaded documents must not automatically be labelled verified. Verification requires an actual authoritative verification process.

10. PRACTICAL USE CASES AND USER JOURNEYS

The following original use cases are retained and expanded.

10.1 Student Example: Future-Proof Technology Career

User request

"I want a future-proof career in technology."

HDOS process

1. Collect education and personal development information.
2. Identify interests and existing skills.
3. Analyze relevant occupations.
4. Compare education programmes.
5. Identify technical and professional skills.

6. Discover certifications and training.
7. Identify internships and scholarships.
8. Generate a development roadmap.

Expected outputs

- Recommended professions.
- Education pathways.
- Required skills.
- Recommended certifications.
- Internship opportunities.
- Earnings information where reliable data exists.
- Five-year development roadmap.

10.2 Graduate Example: Improve Employability

User request

"I want to improve my employability."

HDOS outputs

- Skills-gap analysis.
- Recommended training.
- Market opportunities.
- Job-readiness roadmap.
- CV improvement.
- Interview preparation.
- Internship and vacancy matching.

10.3 Entrepreneur Example: Start a Business

User request

"I want to start a business."

HDOS outputs

- Opportunity assessment.
- Sector recommendations.
- Market research requirements.
- Customer and competitor analysis.
- Startup roadmap.
- Business risk analysis.
- Skills and resource requirements.

- Financing opportunities where verified.

10.4 Citizen Example: Increase Income

User request

"I want to increase my income."

HDOS outputs

- Professional pathways.
- Training opportunities.
- Business opportunities.
- Skills-development recommendations.
- Career-growth roadmap.
- Practical next actions.

10.5 Scholarship Applicant Example

User request

"I want to study abroad and need a scholarship."

HDOS process

1. Analyze the user's academic background.
2. Identify relevant countries and degree levels.
3. Search official scholarship announcements.
4. Compare eligibility criteria.
5. Check academic and document requirements.
6. Identify missing requirements.
7. Prepare application documents.
8. Display the official application channel.
9. Request user approval before submission.
10. Track the application outcome.

10.6 Original Hackathon demonstration example

The original proposal describes a 19-year-old user from Musanze with an interest in technology and the goal of becoming an AI Engineer.

HDOS should demonstrate:

- Skills recommendations.

- Courses and education pathways.
- Practical project recommendations.
- Income information where a reliable source exists.
- Future workforce information.
- A development timeline.

The demonstration should show how user information, actual datasets, rules and AI combine to produce a practical personalized result.

11. NATIONAL DATA ECOSYSTEM AND RWANDA SEED DATA STRATEGY

11.1 Data-driven foundation

The original project identifies four main data groups:

1. NISR Data.
2. Government Data.
3. Open Public Datasets.
4. Educational Data.

This edition expands those groups into an organized data-source ecosystem.

The objective is to connect verified information with the relevant HDOS module.

11.2 Priority data source: NISR

The National Institute of Statistics of Rwanda provides national statistical information.

Potential datasets:

- Population statistics.
- Labour Force Surveys.
- Household living conditions.
- Poverty indicators.
- Education statistics.
- Employment statistics.
- Income and expenditure.
- Sector statistics.
- Demographic indicators.
- Socioeconomic indicators.

Priority imports

Labour Force Survey

Use employment, unemployment, labour-force participation, employment-to-population ratio, labour underutilization, sector and demographic breakdowns.

Rwanda Statistical Yearbook

Use its education, labour, population, business, trade, income, health, agriculture and other sector indicators.

EICV7

Use aggregated poverty, inequality, employment, education, health, housing, living-conditions and socioeconomic information.

Important: statistical observations must remain attached to their reference period and methodology. Data retrieval date is not the same as the period described by the data.

11.3 Priority data source: Ministry of Education

Use the Education Statistical Yearbook and other official information to build an education reference catalogue.

Potential data:

- Education levels.
- School enrolment.
- TVET participation.
- Higher education.
- Polytechnics.
- Education infrastructure.
- Teaching staff.
- Education indicators.
- School-system statistics.

These records support education intelligence and provide national context. They do not automatically provide every institution's current admission requirements or course availability.

11.4 Priority data source: Higher Education Council

The Higher Education Council is an important source for:

- Accredited institutions.

- Academic programme information.
- Higher education announcements.
- Scholarship announcements.
- Study-loan information.
- Academic equivalence information.
- Higher education policies and guidelines.

HDOS should distinguish accredited institutions and programmes from unverified course listings.

11.5 Priority data source: Rwanda Labour Market Information System

Rwanda LMIS is particularly important because its published information covers:

- Occupations.
- Education and training requirements.
- Labour-market trends.
- Skills demand.
- Active labour-market programmes.
- Career guidance.
- Job access and vacancies analysis.
- Sector skills reports.
- Graduate tracer studies.

Use LMIS information as an important foundation for Workforce Intelligence and the Skills Gap Engine.

11.6 Priority data source: Kora Job Portal

Kora supports job discovery, profiles and job matching.

HDOS should complement such existing platforms by using authorized links, published information or permitted integrations.

Potential opportunity fields:

- Job title.
- Employer.
- Location.
- Contract type.
- Required experience.
- Required skills.
- Education.
- Application deadline.

- Original application URL.
- Source and verification date.

HDOS must not assume it has permission to scrape or redistribute information from another platform.

11.7 Priority data source: Internship Rwanda

Potential information includes:

- Internship title.
- Organization.
- Location.
- Required education.
- Skills.
- Duration.
- Eligibility.
- Application procedure.
- Deadline.
- Original link.

11.8 Priority data source: Rwanda Development Board

Use investment opportunity and sector information to support HDOS Business and Income Intelligence.

Potential sectors:

- Agriculture.
- Education.
- Energy.
- Financial services.
- ICT.
- Infrastructure.
- Manufacturing.
- Mining.
- Real estate and construction.
- Tourism.
- Health services.
- Global business services.

A sector investment opportunity is not automatically a grant, scholarship or available job. HDOS must classify these records correctly.

11.9 Seed-data priority catalogue

Data area	Initial source	Main HDOS modules	
Labour indicators	NISR LFS	Workforce, Decision, National Dashboard	
National statistics	NISR Statistical Yearbook	Workforce, Business, Income	
Household and poverty indicators	NISR EICV7	Income, National Dashboard	
Education statistics	MINEDUC	Goal, Education, Decision	
Accredited programmes	HEC	Opportunity, Education, Application	
Occupations	LMIS	Workforce, Career, Skills Gap	
Occupation skills	LMIS and approved occupational sources	Workforce, Skills Gap	
Jobs	Kora and authorized employers	Opportunity, Application	
Internships	Internship Rwanda and employers	Opportunity, Application	
Scholarships	HEC and official scholarship providers	Opportunity, Application	
Business sectors	RDB	Business, Income, Future	
Future skills	LMIS reports and credible research	Future Engine	

11.10 Live data snapshot for the revised proposal

The original proposal did not contain a fully specified and time-stamped national data table. The following snapshot demonstrates how HDOS should use more recent official information.

NISR Labour Force Survey — 2026 Q3

Reference period: August 2026.

Publication/update date shown on NISR's Q3 page: 20 October 2026.

Selected indicators:

Indicator	Published value	Reference	
National unemployment rate	15.6%	August 2026	
Youth unemployment rate	18.8%	August 2026	
Labour-force participation rate	63.9%	August 2026	

Employment-to-population ratio	54.0%	August 2026	
Labour underutilization rate	61.8%	August 2026	

The 2026 Q3 report also shows differences by sex and between rural and urban areas. The system should import the complete official tables rather than relying only on headline statistics.

NISR Labour Force Survey — Annual 2025

Indicator	Published value	Reference	
Unemployment rate	12.4%	Full year 2025	
Employment-to-population ratio	55.9%	Full year 2025	
Services-sector share of employment	44.4%	Full year 2025	

The annual 2025 data and the August 2026 Q3 data describe different reference periods and must not be combined as if they were the same measurement.

MINEDUC Education Statistical Yearbook 2024/2025

Selected published figures include:

Indicator	Published value	Reference	
Total learners in Rwanda	4,822,507	2024/2025	
TVET learners, levels L3–L5	125,653	2024/2025	
Higher education learners	153,318	2024/2025	

NISR EICV7

The seventh Integrated Household Living Conditions Survey covers October 2023 to October 2024. Under the revised methodology described by NISR, the reported poverty rate for 2024 is 27.4%.

These figures are examples of real seed indicators. HDOS should import the official tables and accompanying metadata, not manually hardcode these values as permanent constants.

12. DATA ENGINEERING, QUALITY AND REFRESH ARCHITECTURE

12.1 Data ingestion pipeline

HDOS should use a controlled data pipeline:

DISCOVER → COLLECT → EXTRACT → VALIDATE → NORMALIZE → STORE → PUBLISH → MONITOR → REFRESH

Stage 1: Discover

Identify an official information source and its update mechanism.

Stage 2: Collect

Collect information through an official API, downloadable CSV, Excel, PDF or permitted data interface.

Stage 3: Extract

Convert downloaded information into structured records.

Stage 4: Validate

Check field completeness, dates, numeric ranges, duplicate records and source reliability.

Stage 5: Normalize

Convert data into consistent categories, units, district references, occupation codes and education levels.

Stage 6: Store

Save records together with source information and reference periods.

Stage 7: Publish

Only approved records should enter the production recommendation engine.

Stage 8: Monitor

Identify expired opportunities, broken links and old datasets.

Stage 9: Refresh

Import new information and retain previous versions where historical comparison is useful.

12.2 Data provenance requirements

Every national statistic and external opportunity should contain:

- Source organization.
- Source dataset or announcement.
- Source URL.
- Publication date.
- Reference period.
- Retrieval date.
- Geographic coverage.
- Unit of measurement.
- Methodology where available.
- Verification status.
- Licence or usage conditions where applicable.

12.3 Suggested verification statuses

Status	Meaning	
VERIFIED_OFFICIAL	Confirmed against the authoritative source	
OFFICIAL_REQUIRES_REVIEW	From an official source but not fully checked	
SECONDARY_SOURCE	From a secondary publisher	
USER_PROVIDED	Supplied by the user	
AI_EXTRACTED_UNCONFIRMED	Extracted by AI but not confirmed	
EXPIRED	No longer active	
DISPUTED	Conflicting or uncertain information	
SAMPLE_DATA	Demonstration-only record	

12.4 Data freshness policy

Suggested operational targets:

Dataset	Refresh target	
Jobs	Daily	
Internships	Daily	
Scholarships	Daily	
Grants and competitions	Daily	
Education programmes	On official update and admissions cycle	
Occupation information	Quarterly or on official update	
Skills reports	On publication	
Labour statistics	On official release	
National statistics	On official release	
Goal templates	Quarterly review	

These are HDOS operational targets, not guaranteed publication schedules of the source institutions.

13. DECISION INTELLIGENCE AND PATHWAY FIT INDEX

13.1 Purpose

The Pathway Fit Index helps users compare education, career, skill-development and opportunity options.

It should be transparent, repeatable and based on defined inputs.

13.2 Proposed scoring factors

The following is a proposed initial model, subject to testing and validation.

Criterion	Illustrative weight
Skills alignment	25%
Education and qualification fit	20%
Personal goal and interest alignment	15%
Evidence of labour-market demand	15%
Financial affordability	10%
Time and practical feasibility	10%

Geographic or access suitability	5%
Total	**100%**

A score can be calculated as a weighted combination of normalized factor scores.

For example:

Pathway Fit Index = Σ (Factor Score × Factor Weight)

Each factor should have documented input data and a clear calculation method.

13.3 Scoring interpretation

A suggested presentation:

- 85–100: Strong recorded alignment.
- 70–84: Good recorded alignment.
- 50–69: Partial alignment; development may be needed.
- Below 50: Significant gaps or constraints identified.

These bands are proposed interface conventions, not validated measures of success.

13.4 Explainability requirements

Each score must show:

- Factors considered.
- Individual factor scores.
- Weights.
- Missing information.
- Relevant sources.
- Date of evaluation.
- Algorithm version.
- Limitations.

13.5 Eligibility must be separate from ranking

A recommendation score must not be confused with official opportunity eligibility.

Eligibility should be evaluated using the actual published requirements.

For example:

- ✓ Nationality requirement.
- ✓ Age requirement.
- ✓ Academic qualification.
- ✓ Minimum academic result.
- ✓ Language requirement.
- ✓ Relevant experience.
- ✓ Required documents.
- ✓ Application deadline.

A user may receive a high compatibility score but still fail one mandatory eligibility condition.

The system must explain this clearly.

14. ARTIFICIAL INTELLIGENCE AND RECOMMENDATION ARCHITECTURE

14.1 AI's role

AI should support the interpretation of evidence and help generate personalized guidance.

Possible uses:

- Natural-language goal understanding.

1. Profile summarization.
2. Skills-gap explanations.
3. Career pathway explanations.
4. Learning-roadmap generation.
5. Document extraction.
6. CV and motivation-letter assistance.
7. Interview preparation.
8. Conversational support.
9. Report generation.

14.2 Rules engine and AI engine

The platform should separate deterministic logic from generative AI.

Rules-based intelligence:

- Eligibility conditions.

- Required qualifications.
- Deadline status.
- Mandatory documents.
- Basic scoring rules.
- Service authorization.
- Credit deductions.

AI-supported intelligence:

- Explaining results.
- Summarizing documents.
- Drafting application materials.
- Comparing options in natural language.
- Producing development suggestions.
- Helping users understand complex requirements.

AI should not be allowed to secretly alter eligibility rules or invent official opportunity requirements.

14.3 Source-grounded responses

Where HDOS provides a factual recommendation, it should identify the information source.

AI should:

1. Retrieve relevant verified information.
2. Use the available source content.
3. Explain what the evidence supports.
4. identify uncertainty and missing information.
5. Avoid creating false facts.
6. Give links to original publishers when available.

14.4 Model integration

The original proposal identifies OpenAI APIs as part of the AI layer.

The implementation may use one or more approved AI providers, subject to cost, security, availability, quality and data-protection requirements.

Provider changes must not change the meaning of official eligibility rules or deterministic scores.

14.5 Human responsibility

HDOS is a decision-support platform. Users remain responsible for major personal decisions and official declarations.

Professional, institutional or legal advice must be identified as such and not represented as an automatic AI conclusion.

15. HDOS APPLICATION PORTAL

15.1 Strategic purpose

The Application Portal converts opportunity discovery into organized application action.

It connects Opportunity Intelligence, Life DNA, Decision Intelligence and the Document Vault.

15.2 Main components

Applicant Profile

Stores:

- Contact details.
- Education history.
- Qualifications.
- Skills.
- Experience.
- Languages.
- Preferences.
- Application information.

Opportunity Explorer

Displays:

- ✓ Opportunity title.
- ✓ Provider.
- ✓ Category.
- ✓ Location.
- ✓ Qualifications.
- ✓ Skills.
- ✓ Eligibility.
- ✓ Deadline.
- ✓ Benefits.
- ✓ Official application link.
- ✓ Verification date.

Eligibility Engine

Compares the applicant's profile against mandatory and optional requirements.

Document Matching

Matches the required documents with the user's stored documents.

Application Preparation

Supports:

- CV preparation.
- Motivation letters.
- Cover letters.
- Personal statements.
- Supporting document checklists.
- Application-form preparation.

Application Dashboard

Tracks:

i. Selected opportunity.
ii. Application status.
iii. Missing information.
iv. Deadline.
v. Review tasks.
vi. Submission date.
vii. Confirmation.
viii. Follow-up actions.

15.3 Application lifecycle

Suggested statuses:

DISCOVERED
ELIGIBILITY_CHECK
ELIGIBLE
INELIGIBLE
DOCUMENTS_REQUIRED
PREPARING
READY_FOR_REVIEW
APPROVED
SUBMITTING

SUBMITTED
UNDER_REVIEW
INTERVIEW
ACCEPTED
REJECTED
WITHDRAWN
EXPIRED
ARCHIVED

ARCHIVED is a lifecycle state. It must not overwrite the actual outcome.

15.4 Application automation policy

HDOS should support three levels:

Level 1 — Preparation

The system prepares information and documents for the user's review.

Level 2 — User-approved submission

HDOS submits through an approved integration after the user reviews and explicitly approves the final application.

Level 3 — Limited pre-authorized submission

Only permitted where official systems and provider integrations support it, the user has provided clear authorization and no personal declaration or restricted action requires separate confirmation.

HDOS must not bypass CAPTCHA, account protections, application restrictions or legal terms.

16. HDOS SECURE DOCUMENT VAULT

16.1 Purpose

The Document Vault gives users one secure location for documents used in educational, employment, training and scholarship applications.

16.2 Initial document catalogue

Document	Category	
CV or Résumé	Professional	
Diploma	Academic	
Academic Transcript	Academic	
Recommendation Letter	Supporting	
Passport	Identity	
National ID	Identity	
Certificates	Qualification	
Cover Letter	Application	
Motivation Letter	Application	
Portfolio	Professional	
Language Certificate	Qualification	
Proof of Address	Supporting	
Professional Licence	Qualification	
Other Supporting Documents	Supporting	

16.3 Document processing

A document may pass through:

UPLOADED → PROCESSING → EXTRACTED → AWAITING_USER_CONFIRMATION → CONFIRMED

Where actual authoritative verification exists, additional states may be used:

VERIFIED, REJECTED or EXPIRED.

Extraction is not verification.

A scanned diploma that has been read by AI is not an officially verified diploma.

16.4 Document security

Requirements:

- Private storage.
- Encryption in transit and at rest.
- Authorized access.
- Secure download links.
- Malware scanning.
- Audit logs.
- File type and size validation.
- Retention controls.
- Secure deletion.
- Document versioning.
- Appropriate backups.

16.5 Opportunity-specific document readiness

Before an application is marked ready, HDOS should:

1. Read official document requirements.
2. Identify mandatory documents.
3. Match available documents.
4. Check expiry where relevant.
5. Identify missing documents.
6. Request user confirmation.
7. Preserve the exact document versions used.

17. HDOS VALUE AND INTELLIGENCE ECONOMY

17.1 Commercial principle

HDOS should not operate only as a subscription-collection system.

It should be a service-consumption platform in which users pay for actual intelligence, reports, processing and assistance they choose to consume.

This model connects the platform's revenue to delivered customer value.

17.2 HDOS Value and Intelligence Economy™

The proposed component consists of:

1. Service Catalogue.

2. Pricing Engine.
3. Usage Metering.
4. User Wallet.
5. Credit Management.
6. Subscription Management.
7. Payment Gateway.
8. Billing and Invoicing.
9. Refund Management.
10. Revenue Analytics.

17.3 Service catalogue

Service	Deliverable	Proposed charging model	
Goal Intelligence	Structured goal assessment and roadmap	Per assessment	
Life DNA Analysis	Personal profile analysis	Per analysis	
Career Intelligence	Career fit and skill-gap report	Per report	
Workforce Intelligence	Detailed occupation or sector report	Per report	
Advanced Opportunity Matching	Personalized opportunity comparison	Usage-based	
Scholarship Eligibility Assessment	Requirements comparison and eligibility report	Per assessment	
CV Improvement	Improved CV draft	Per document	
Motivation Letter Assistance	Opportunity-specific letter draft	Per document	
Interview Preparation	Practice and feedback session	Per session	
Document Intelligence	Document extraction and classification	Per processed document	
Full Application Preparation	Coordinated application package	Per package	
Institutional Intelligence	Aggregated institutional reporting	Institutional licence	
API Intelligence	Authorized programmatic intelligence access	Usage-based contract	

Public opportunity discovery should remain free or affordable. Users should not be forced to purchase a subscription simply to see a public job or scholarship.

17.4 Three charging models

A. Pay-per-service

The user pays for one selected service.

Example:

A customer requests a detailed scholarship eligibility assessment. HDOS displays the service description, fee, expected deliverables and limitations before payment.

B. Prepaid intelligence credits

Users purchase a package of credits and consume them when they request billable services.

Illustrative consumption catalogue:

Service	Example credit requirement	
Detailed Goal Assessment	5	
Life DNA Analysis	5	
Career Pathway Report	8	
Scholarship Eligibility Assessment	4	
CV Improvement	5	
Motivation Letter Assistance	5	
Interview Preparation	6	
Full Application Preparation	15	

These are design examples for testing. Final credit prices must be based on operational costs and customer research.

C. Subscription with included usage

Subscriptions provide access to a defined allowance of services.

The user may purchase additional credits after using the included allowance.

17.5 Proposed service plans

Plan	Proposed price	Main structure	
Free	RWF 0	Basic profile, goals and public opportunity discovery	
Student Plus	RWF 2,500/month	Additional intelligence allowance	
Student Premium	RWF 5,000/month	Higher allowance and advanced services	
Student Annual	RWF 15,000/year	Annual student access package	

Application Assistance	RWF 5,000–15,000	Service package depending on complexity	
Institution Starter	RWF 300,000/month	Institutional access and reporting	
Institution Professional	RWF 750,000/month	Expanded institutional services	
Institution Enterprise	From RWF 1,500,000/month	Custom integrations and support	
Employer Licence	From RWF 500,000/month	Workforce intelligence services	
Custom Deployment	Quotation-based	Agreed project scope	

These are proposed commercial hypotheses, not confirmed market prices. They require willingness-to-pay research, unit-cost calculation, pilot testing and review before publication.

17.6 Usage and credit control

A paid service should follow:

1. User selects a service.
2. HDOS displays the deliverable and exact price.
3. The system checks available credits or payment status.
4. Credits are reserved.
5. Service execution is authorized.
6. Actual usage is recorded.
7. Successful service delivery settles the reserved credits.
8. Eligible failed operations release or refund the reservation.
9. A receipt and service history are generated.

The user must be able to see their available balance, service history, prices and spending limits.

17.7 Responsible AI consumption

AI token usage should help HDOS calculate internal costs. It should not be the only customer-facing pricing measure.

Users understand completed services and deliverables better than internal model-token counts.

A user should know whether a service includes one report, multiple revisions, a human review or additional work.

18. BUSINESS MODEL, REVENUE AND PAYMENT INFRASTRUCTURE

18.1 Main revenue streams

HDOS may generate revenue through:

- Individual subscriptions.
- Prepaid intelligence credits.
- Pay-per-service assessments.
- Application preparation.
- Document intelligence.
- Institutional licensing.
- Employer and enterprise services.
- Authorized API integrations.
- Custom institutional deployments.
- Training and consultancy.

18.2 Freemium strategy

The basic platform should offer meaningful free value.

Recommended free services:

- ✓ Account creation.
- ✓ Basic personal profile.
- ✓ Goal creation.
- ✓ Public opportunity discovery.
- ✓ Basic career and education exploration.
- ✓ Basic development guidance.

Advanced paid services may include:

- Detailed analysis.
- Advanced comparisons.
- Extensive reports.
- Premium AI assistance.
- Application preparation.
- Additional document processing.
- Institutional analytics.

18.3 Rwanda payment channels

Potential payment methods:

- MTN Mobile Money.
- Airtel Money.
- Visa.
- Mastercard.
- Bank transfer.
- Other locally supported digital payment methods.

Potential payment providers should be evaluated on:

- Merchant onboarding.
- API support.
- Transaction costs.
- Webhooks.
- Settlement.
- Refund processing.
- Recurring-payment support.
- Customer support.
- Security and compliance.

Potential providers to evaluate include Mocoard, UrubutoPay, Flutterwave, DPO Pay and other eligible Rwanda payment-service providers.

Their availability, fees, production approval and contractual terms must be confirmed before a live integration is selected.

18.4 Payment lifecycle

SERVICE SELECTION → PRICE CONFIRMATION → PAYMENT REQUEST → PROVIDER AUTHORIZATION → SERVER VERIFICATION → SERVICE EXECUTION → USAGE SETTLEMENT → RECEIPT

The application backend must verify transactions directly with the payment provider or through authenticated webhooks.

A successful browser redirect alone must never activate a paid service.

18.5 Billing requirements

The payment subsystem should support:

- Unique transaction references.
- Unique invoice numbers.
- Payment-provider references.

- Payment status.
- Payment timestamps.
- Currency.
- Gross amount.
- Fees.
- Net settlement.
- Refunds.
- Credit adjustments.
- Subscription renewals.
- Payment reconciliation.
- Audit history.

18.6 Illustrative financial model

The following is a planning scenario, not actual revenue or a guaranteed forecast.

Illustrative monthly revenue

Revenue source	Quantity	Unit price
Student Plus	500	RWF 2,500
Institutions	5	RWF 1,200,000
Enterprise licences	3	RWF 500,000
Application support	50	RWF 10,000
Total		

Illustrative monthly operating costs

Cost area	Amount	
Engineering and technical team	RWF 2,800,000	
Cloud, AI and infrastructure	RWF 700,000	
Marketing and sales	RWF 500,000	
Support and administration	RWF 400,000	
Compliance and contingency	RWF 400,000	
Total	**RWF 4,800,000**	

Illustrative monthly operating surplus:

RWF 9,250,000 − RWF 4,800,000 = RWF 4,450,000

Illustrative annualized amounts:

- Revenue: RWF 111,000,000.
- Operating costs: RWF 57,600,000.
- Operating surplus: RWF 53,400,000.

These assumptions do not account for every potential expense, tax, financing, depreciation, customer acquisition cost, payment fee or collection delay. They are not a first-year financial forecast and must be tested against realistic customer acquisition and retention data.

19. TECHNICAL AND SOFTWARE ARCHITECTURE

19.1 Original architecture

The original proposal describes:

- HDOS Web Platform.
- HDOS Mobile App.
- API Gateway.
- HDOS AI Engine.
- Goal Engine.
- Life DNA.
- Workforce Intelligence.
- Opportunity Intelligence.
- Decision Intelligence.
- Future Engine.
- Data Platform.
- NISR datasets.
- Education data.
- Workforce data.
- Economic data.
- Open datasets.

This architecture remains the strategic foundation.

19.2 Original technology roadmap

The original proposal identifies the following technologies for Version 1.0:

Frontend

- Next.js.

- React.
- TypeScript.

Backend

- Node.js.
- Express.
- TypeScript.

Database

- PostgreSQL.

AI Layer

- OpenAI APIs.

Recommendation Engine

- Rules-based intelligence.

These are the original proposed technical choices. The actual implementation must be inspected before they are described as confirmed current stack components.

19.3 Expanded architecture

Recommended logical architecture:

USER INTERFACE

Web application and future mobile application.

↓

API GATEWAY

Authentication, authorization, request validation and rate limits.

↓

CORE APPLICATION SERVICES

- ✓ User and Profile Service.
- ✓ Goal Service.
- ✓ Life DNA Service.

- ✓ Education Intelligence Service.
- ✓ Workforce Intelligence Service.
- ✓ Opportunity Service.
- ✓ Decision Service.
- ✓ Application Service.
- ✓ Document Service.
- ✓ Billing and Wallet Service.
- ✓ Notification Service.

↓

INTELLIGENCE ORCHESTRATION

- ✓ Rules Engine.
- ✓ Pathway Fit Index.
- ✓ Eligibility Engine.
- ✓ AI Service.
- ✓ Report Generation.
- ✓ Usage Metering.

↓

DATA PLATFORM

- ✓ Structured database.
- ✓ Search index.
- ✓ Document storage.
- ✓ Statistical data repository.
- ✓ Data-source registry.
- ✓ Historical data store.

↓

EXTERNAL INTEGRATIONS

- ✓ NISR and official statistical sources.
- ✓ MINEDUC.
- ✓ HEC.
- ✓ LMIS.
- ✓ Opportunity publishers.
- ✓ Payment providers.
- ✓ Approved AI providers.

19.4 Technical evolution

Version 1.0 — Rules-Based Intelligence

- Core user profiles.
- Goals.
- Education information.
- Occupations.
- Opportunity discovery.
- Explainable recommendations.
- Basic roadmap generation.

Version 2.0 — Predictive Intelligence

- Machine learning.
- Predictive models.
- Recommendation ranking.
- Skills scoring.
- Better historical analysis.

Version 3.0 — Knowledge Intelligence

- Knowledge graph.
- Behaviour intelligence with appropriate privacy controls.
- Adaptive guidance.
- Entity relationships.

Version 4.0 — Advanced Decision Intelligence

- Multi-goal optimization.
- Advanced decision support.
- Scenario analysis.
- More adaptive goal planning.

Predictive features should only be activated after data availability, quality, evaluation and fairness requirements are satisfied.

20. DATABASE ARCHITECTURE AND SEED DATA MODEL

20.1 Database entities

Recommended logical entities:

Entity	Purpose

data_sources	Records source publishers and source metadata
data_import_runs	Tracks every data import
districts	Administrative geographic reference data
education_levels	Education classification
institutions	Education and training providers
academic_programmes	Academic and training programmes
skills	Technical, professional and transferable skills
occupations	Occupation catalogue
occupation_skills	Skills associated with occupations
labour_indicators	Statistical indicators
opportunities	Jobs, scholarships, internships and programmes
opportunity_requirements	Eligibility and application conditions
goal_templates	Reusable development goals
pathway_templates	Reusable education and career pathways
eligibility_rules	Deterministic requirements
service_catalogue	Paid and free HDOS services
service_prices	Version-controlled prices
wallets	User credit balances
wallet_transactions	Credit movement ledger
payments	External payment records
invoices	Billing records
service_usage	Actual service consumption
user_profiles	User profile information
user_skills	User skills
user_goals	Personal goals
user_applications	Application tracking
document_categories	Document classification
applicant_documents	User-owned documents
document_versions	Document history
document_extractions	AI-extracted fields
document_verifications	Actual verification records
application_documents	Documents used in applications
audit_logs	Security and activity records

20.2 Data record example

Every indicator should preserve source information.

```
{
  "indicator_code": "UNEMPLOYMENT_RATE",
  "value": 15.6,
  "unit": "percent",
  "geography": "Rwanda",
  "reference_period": "2026-Q3",
  "period_description": "August 2026",
  "source_organization": "NISR",
  "source_dataset": "Labour Force Survey Q3 2026",
  "source_url": "https://www.statistics.gov.rw/data-sources/surveys/Labour-
Force-Survey/labour-force-survey-2026/labour-force-survey-2026-q3",
  "published_at": "2026-10-20",
  "retrieved_at": "2026-11-03",
  "verification_status": "VERIFIED_OFFICIAL",
  "is_demo": false
}
```

The date and value in this example describe a specific official statistical publication. The system should import the full dataset and not hardcode the indicator permanently.

20.3 Occupation seed record example

```
{
  "occupation_code": "ICT_AI_ENGINEER",
  "title": "Artificial Intelligence Engineer",
  "sector": "Information and Communication Technology",
  "description": "Develops and deploys AI-powered software systems.",
  "skills": [
    "Python",
    "Statistics",
    "Machine Learning",
    "Data Engineering",
    "Software Development"
  ],
  "education_pathways": [
    "Computer Science",
    "Software Engineering",
    "Information Technology",
    "Data Science"
  ],
  "source_type": "OCCUPATIONAL_REFERENCE",
  "source_url": null,
  "verification_status": "REQUIRES_SOURCE_VALIDATION",
  "is_demo": false
}
```

This is a sample structure. Specific skills and education pathways must be validated against official occupation information and relevant employers or institutions.

20.4 Scholarship seed record example

```
{
  "title": "Government Study Loan",
  "provider": "Government of Rwanda",
  "category": "STUDENT_FINANCIAL_SUPPORT",
  "intake": "JANUARY_2027",
  "reference_source": "Higher Education Council",
  "official_announcement_url": "https://www.hec.gov.rw/updates/scholarships",
  "eligibility_rules": [],
  "required_documents": [],
  "deadline": null,
  "status": "REQUIRES_ANNOUNCEMENT_REVIEW",
  "verification_status": "OFFICIAL_REQUIRES_REVIEW"
}
```

The system must extract the actual requirements and deadline from the specific announcement before presenting them as confirmed.

20.5 Goal pathway example

```
{
  "goal_title": "Become an AI Engineer",
  "goal_category": "CAREER",
  "milestones": [
    "Assess mathematics and programming knowledge",
    "Learn Python programming",
    "Study statistics and machine learning",
    "Build practical AI projects",
    "Develop a professional portfolio",
    "Identify relevant internships and jobs"
  ],
  "source_type": "HDOS_GENERATED_PATHWAY",
  "is_official_qualification": false
}
```

21. SECURITY, PRIVACY AND RESPONSIBLE AI

21.1 Personal information

HDOS may process information such as:

- Name and contact details.
- Education history.
- Skills and experience.
- Goals.
- Application documents.
- Identity documents.

- Payment records.
- User-generated information.

This creates privacy and security responsibilities.

21.2 Rwanda legal framework

The platform must assess and comply with Rwanda's Law No. 058/2021 of 13 October 2021 relating to the protection of personal data and privacy, together with applicable regulations and guidance.

The project must obtain appropriate legal and data-protection advice before production deployment.

21.3 Required controls

- ✓ Appropriate registration and compliance review.
- ✓ Clear privacy notices.
- ✓ Lawful processing.
- ✓ Purpose limitation.
- ✓ Data minimization.
- ✓ Access control.
- ✓ Encryption.
- ✓ Audit logging.
- ✓ Secure retention and deletion.
- ✓ Processor contracts where required.
- ✓ Security incident response.
- ✓ Data-protection impact assessments where applicable.
- ✓ Controlled international data transfers.
- ✓ User rights and support processes.

21.4 Automated profiling and decision support

HDOS may perform automated profile analysis and recommendations.

For consequential recommendations, it should explain the logic, relevant input categories and likely effects.

A recommendation should not automatically deny a person's opportunity to study or work based only on an opaque AI score.

21.5 AI safety

The AI system must not:

1. Invent a scholarship.
2. Fabricate an institution's accreditation.
3. Create false job vacancies.
4. Make up application requirements.
5. Claim an applicant is officially eligible without sufficient evidence.
6. Falsely claim a document has been verified.
7. Guarantee employment, admission, grants or income.
8. Hide material limitations.

22. USER INTERFACE AND PRODUCT EXPERIENCE

22.1 Public website

Recommended pages:

1. Home.
2. About HDOS.
3. How It Works.
4. Intelligence Modules.
5. Opportunities.
6. Application Portal.
7. Data and Methodology.
8. Pricing and Services.
9. Contact and Partnerships.
10. Sign In.
11. Create Account.

22.2 Authenticated dashboard

Recommended navigation:

- Command Centre.
- Goals and Roadmap.
- Life DNA.
- Workforce Insights.
- Opportunities.
- Decision Assistant.
- Future Engine.
- Applications.
- Document Vault.

- My Wallet and Services.
- Notifications.
- Settings.

22.3 Command Centre

The home dashboard should help the user understand:

- Current main goal.
- Progress and milestones.
- Recommended next actions.
- Relevant opportunities.
- Skills gaps.
- Active applications.
- Important deadlines.
- Document readiness.
- Service allowance and wallet balance.

22.4 My Wallet and Services

The payment and service area should show:

- Available balance.
- Available credits.
- Free and paid services.
- Service descriptions.
- Exact prices.
- Service usage history.
- Purchase credits.
- Invoices and receipts.
- Refund records.
- Subscription management.
- Spending limits.
- Payment preferences.

22.5 Experience standards

The product should support:

- Desktop, tablet and mobile layouts.
- Responsive navigation.
- English and future Kinyarwanda localization.
- Accessible forms.
- Keyboard navigation.
- Screen-reader compatibility.

- Low-bandwidth use.
- Clear loading and error states.
- Useful empty states.
- Secure authentication.
- Visible data source dates.
- Consistent design components.

23. COMPETITIVE ADVANTAGE AND INNOVATION

23.1 Existing systems are important complements

HDOS should not claim that existing services do not provide career matching, workforce information or scholarship announcements.

For example:

- Rwanda LMIS provides labour-market and occupational intelligence.
- Kora supports job access and job matching.
- HEC provides scholarship and higher-education information.
- Internship Rwanda supports professional internship access.
- NISR provides statistical evidence.

HDOS can use these existing public information sources as part of its integrated intelligence model, subject to their data access and usage rules.

23.2 Proposed differentiation

HDOS aims to connect information that is commonly accessed through separate services into a single goal-oriented workflow.

Its differentiation is the integration of:

1. Personal goals.
2. Personal development profiles.
3. National statistics.
4. Education pathways.
5. Workforce intelligence.
6. Opportunity discovery.
7. Explainable fit assessment.
8. Application preparation.

9. Document management.
10. Service consumption and value-based billing.
11. Progress tracking.
12. Outcome evaluation.

23.3 Competitive positioning

Most platforms focus on a particular information or service category. HDOS is designed to organize the broader human-development decision journey around the individual.

That is the strategic proposition to validate through the hackathon prototype, pilot users and institutional feedback.

24. NATIONAL DEVELOPMENT ALIGNMENT

24.1 Alignment with NST2

The original proposal connects HDOS to the Second National Strategy for Transformation.

HDOS can contribute to relevant national priorities through:

Human Capital Development

By improving access to skills, education and career intelligence.

Youth Employment

By supporting skills development, job readiness and access to opportunities.

Education Quality and Relevance

By helping users compare education pathways with occupational requirements.

Digital Transformation

By applying AI and data systems to human-development decision support.

Innovation

By demonstrating an integrated approach to evidence-based personal development.

Private Sector Growth

By improving access to workforce and entrepreneurship intelligence.

HDOS is a proposed contributor to these objectives. It should not claim direct responsibility for achieving national outcome targets.

24.2 Alignment with Vision 2050

The original proposal identifies the following areas:

- Knowledge-based economy development.
- Human capital transformation.
- Innovation-driven growth.
- Digital economy development.
- Productivity enhancement.
- Poverty reduction.

HDOS supports these ambitions by developing accessible intelligence services that help people make better-informed decisions.

25. IMPLEMENTATION AND COMMERCIALIZATION ROADMAP

The original four-phase roadmap is retained.

The dates should be treated as relative to the approved project commencement date and reviewed against available resources.

PHASE 1: HACKATHON MVP

Original timeline: Month 0–3.

Objective

Build a functional prototype demonstrating how statistical and personal information can support practical human-development decisions.

Original core features

User Profile

- Age.
- District.
- Education level.
- Current skills.
- Interests.
- Career goals.
- Income goals.

Education Intelligence

Uses education statistics, skills information and programme information.

Generates:

- Recommended courses.
- Recommended universities.
- Skills roadmap.

Career Intelligence

Uses labour statistics, employment data and workforce trends.

Generates:

- Career matches.
- Demand information.
- Salary estimates only where reliable data is available.
- Required skills.

Opportunity Dashboard

Displays:

- Scholarships.
- Training programmes.
- Internships.
- Business opportunities.

Recommendation Engine

Generates:

- Personalized development plan.
- One-year roadmap.
- Three-year roadmap.
- Five-year roadmap.

Proposed MVP additions

- Real data-source registry.
- Initial occupation catalogue.
- Initial education-programme catalogue.
- Initial opportunity catalogue.
- Transparent Pathway Fit Index.
- Verified data references.
- Basic application tracking.
- Basic service catalogue.

Hackathon architecture

Frontend → React Dashboard → Node.js Backend → HDOS Recommendation Engine → NISR Data Layer and permitted public datasets.

PHASE 2: NATIONAL PILOT

Original timeline: Year 1.

Objective

Pilot HDOS in schools, universities and other education or workforce-development settings.

Original target users

- Secondary school students.
- TVET students.
- University students.
- Graduates.
- Job seekers.

Original new modules

HDOS Learn™

- Learning recommendations.
- Academic roadmaps.
- Certification tracking.

HDOS Career™

- Career planning.
- Job readiness.
- Workforce intelligence.

Skills Gap Engine

- Current skills.
- Market demand.
- Future skills.
- Skills-gap reports.

Proposed additions

- Application Portal.
- Document Vault.
- Scholarship assessment.
- CV and interview assistance.
- Paid service catalogue.
- Wallet and credits.
- First payment integration.
- Institutional onboarding.
- Data-quality monitoring.

Original institutional partnership targets

- RP Colleges.
- Universities.
- TVET Institutions.
- Training Centres.

Any partnership must be formally agreed.

Original pilot metrics

- Students registered.
- Career plans generated.
- Skills assessments completed.
- Recommendations followed.

Additional metrics should measure user satisfaction, data quality, application completion and actual progress.

PHASE 3: NATIONAL DEVELOPMENT PLATFORM

Original timeline: Year 2–Year 4.

Objective

Develop HDOS into a broader national human-capital intelligence platform.

Original HDOS Workforce Intelligence™

Uses:

- Labour Force Surveys.
- Employment statistics.
- Economic indicators.

Produces:

- National skills-gap analysis.
- Employment forecasts.
- Future workforce-demand scenarios.

Original HDOS Business™

Supports:

- Entrepreneurs.
- SMEs.
- Startups.

Provides:

- Business opportunity analysis.
- Market intelligence.
- Entrepreneurship plans.

Original HDOS Income Intelligence™

Uses:

- Income statistics.
- Regional data.
- Sector data.

Provides:

- Income-improvement plans.
- Career-growth recommendations.

Original Government Dashboard

Potential institutional audiences:

- MINEDUC.

- MINICT.
- MIFOTRA.
- NISR.
- Development partners.

Potential capabilities:

- Skills mapping.
- Workforce planning.
- Youth employment tracking.
- Regional opportunity mapping.

Access must be based on formal agreements, legal authority, data minimization and privacy controls.

PHASE 4: GLOBAL HDOS

Original timeline: Year 5–Year 10.

Objective

Expand HDOS beyond Rwanda.

Original target markets

- East Africa.
- Africa.
- Global markets.

Original intelligence layers

Future Skills Intelligence™

- AI careers.
- Green economy.
- Automation.
- Digital transformation.

Global Opportunity Engine™

- Countries.
- Jobs.
- Scholarships.
- Universities.
- Remote work opportunities.

Global Decision Intelligence™

Helps users understand:

- Where should I study?
- Where should I work?
- Which skills should I learn?
- Which opportunities fit me best?

International expansion requires country-specific datasets, legal reviews, localization, institutional partnerships and verification of opportunity sources.

26. TEAM RESPONSIBILITIES AND GOVERNANCE

26.1 IRAVUGA Chrismas

Founder & Chief Systems Architect

Original responsibilities

- Platform vision.
- AI architecture.
- Decision intelligence design.
- Human-development frameworks.
- System strategy.

Phase 1

- Vision.
- System architecture.
- AI logic design.
- Goal Engine design.
- Decision framework.

Phase 2

- Intelligence models.
- Opportunity graph.
- Recommendations.
- User journey design.

Phase 3

- Government intelligence.
- National analytics.
- AI evolution.

Proposed additional responsibilities

- Data architecture.
- Platform security coordination.
- Service orchestration.
- Technical quality assurance.
- Intellectual property strategy.

26.2 RUDASINGWA Bienvenue

Co-Founder & Chief Growth & Business Intelligence Officer

Original responsibilities

- Business strategy.
- Market intelligence.
- Commercial partnerships.
- User acquisition.
- Growth strategy.

Phase 1

- Business model.
- Market research.
- User requirements.
- Competitor analysis.

Phase 2

- Partnerships.
- Pilot programmes.
- Institutional engagement.

Phase 3

- National expansion.
- Commercial strategy.
- Revenue growth.

Proposed additional responsibilities

- Customer research.
- Pricing experiments.
- Institutional service packages.
- Business development.
- Customer success.

26.3 Academic background retained from the original proposal

IRAVUGA Chrismas:

- RP Musanze College.
- Advanced Diploma in Electrical Automation Technology.

RUSAINGWA Bievenue:

- RP Musanze College.
- Advanced Diploma in E-Commerce.

27. SUCCESS METRICS, EVALUATION AND MONITORING

27.1 Original aspirational targets

The original proposal identifies:

Year 1

- 10,000 users.
- 100 career plans.
- 10 pilot institutions.

Year 3

- 100,000 users.
- National adoption.
- Government partnerships.

Year 5

- 1 million users.
- Regional expansion.
- Multiple African countries.

These are aspirational project targets, not achieved results or guaranteed forecasts. They should be reviewed against available resources and realistic adoption rates.

27.2 Improved performance indicators

User engagement

- Registrations.
- Completed profiles.
- Active users.
- Goal creation.
- Returning users.

Intelligence quality

- Recommendation relevance.
- Data freshness.
- Source coverage.
- Explanation completeness.
- User feedback.
- Error and correction rates.

Education outcomes

- Education pathways explored.
- Programmes compared.
- Scholarship opportunities discovered.
- Application readiness.

Employment outcomes

- Job opportunities explored.
- Skills gaps identified.
- Training pathways selected.
- Applications submitted.
- Interviews obtained.
- Employment outcomes voluntarily reported.

Application performance

- Applications prepared.
- Applications submitted.
- Missing-document rate.
- Deadline compliance.
- Application status updates.
- User satisfaction.

Commercial performance

- Paid-service conversion.
- Credits consumed.
- Revenue per active customer.
- Customer retention.
- Institutional contracts.
- Service delivery cost.
- Refund rate.
- Gross margin.

National and institutional impact

- Pilot institutions.
- Number of beneficiaries.
- Regional reach.
- Skills-development participation.
- Institutional reporting usage.

HDOS should measure outcomes without claiming that correlation proves HDOS caused a national employment or poverty change.

28. RISK ANALYSIS AND MITIGATION

Risk	Potential impact	Recommended mitigation
Outdated statistical data	Misleading recommendations	Source dates, refresh schedule and stale-data warnings
Incorrect opportunity records	Users miss deadlines or submit invalid applications	Official-source verification and expiry monitoring
AI hallucinations	False guidance	Retrieval-grounded responses and clear uncertainty
Biased recommendations	Unfair pathway rankings	Transparent rules and periodic fairness evaluation

Privacy breach	Exposure of personal documents	Encryption, authorization, audit logs and incident response
Payment fraud	Financial loss	Provider-side verification and transaction reconciliation
High AI costs	Unsustainable service delivery	Usage metering, caching, model selection and limits
Low customer adoption	Weak commercial sustainability	Pilots, user research and affordable free entry
Dependence on external platforms	Service interruption	Modular connectors and alternative data sources
Incomplete institutional data	Misleading education recommendations	Verification status and source limitations
Unvalidated forecasts	False confidence	Scenario labels and model validation
Unclear legal responsibilities	Regulatory and contractual risk	Legal review and institutional agreements

29. HACKATHON DEMONSTRATION PLAN

29.1 Demonstration objective

Show how HDOS transforms a user's goal and national data into a transparent development pathway and actionable opportunities.

29.2 Demonstration scenario

A 19-year-old student from Musanze wants to become an AI Engineer.

Demonstration sequence

1. Register or open the demonstration profile.
2. Enter education level, skills, interests and goal.
3. Generate a Life DNA profile.
4. Retrieve relevant occupation information.
5. Compare the user with required skills.
6. Display recommended learning pathways.
7. Retrieve education programme information.
8. Show relevant opportunities.
9. Generate a personalized roadmap.

10. Demonstrate an application-readiness checklist.
11. Show document requirements.
12. Display the service catalogue and explain a paid intelligence service.
13. Show the wallet and payment workflow in a clearly labelled sandbox or demonstration mode.
14. Display the final user roadmap and next actions.

29.3 What judges should see

- Real data provenance.
- A working AI capability.
- Transparent scoring.
- Practical recommendations.
- Connected intelligence modules.
- A credible national problem.
- A feasible initial product.
- An expansion strategy.

The demonstration should not show fake live payments, fabricated opportunities or unverified national statistics as if they were actual production transactions.

30. INTELLECTUAL PROPERTY AND FUTURE DEVELOPMENT

30.1 Potential innovation areas

The proposed intellectual-property areas include:

1. Goal-to-opportunity intelligence architecture.
2. Integrated human-development profile.
3. National-data-to-personal-pathway workflow.
4. Explainable pathway comparison.
5. Opportunity eligibility and document matching.
6. Development-progress tracking.
7. AI service-consumption architecture.
8. Integrated application intelligence.

These are candidate areas for technical and legal assessment. Their inclusion in this book does not establish that they are novel, patentable or free from prior art.

30.2 Prior-art and competitive analysis

Before making a patent claim or asserting worldwide uniqueness, the project should conduct:

- Patent searches.
- Academic publication searches.
- Product and platform comparisons.
- Existing national-platform reviews.
- Technical novelty mapping.
- Freedom-to-operate assessment.

Existing platforms such as labour-market systems, job portals, scholarship portals and AI assistants should be treated as relevant prior art or competitive references.

30.3 Protection strategy

Potential forms of protection include:

- Trademark protection for the HDOS name and logos.
- Copyright in original software and documentation.
- Trade-secret protection for confidential implementation methods.
- Patent assessment for any demonstrably novel technical invention.
- Contracts defining ownership of contributed code, datasets and content.

31. CONCLUSION

HDOS Rwanda™ proposes a new integrated approach to human-development decision support.

The original concept identified a major problem: people have access to substantial amounts of information but still struggle to convert it into informed decisions about education, employment, skills, entrepreneurship and economic progress.

The consolidated design retains the original six intelligence modules and extends the platform with practical capabilities for real users.

Through the Goal Engine, Life DNA, Workforce Intelligence, Opportunity Intelligence, Decision Intelligence and Future Engine, HDOS aims to connect personal ambitions with evidence-based development options.

The addition of the Application Portal and Secure Document Vault extends the platform beyond recommendations into practical opportunity execution.

The addition of the Value and Intelligence Economy connects paid services with actual customer consumption and delivered value rather than relying only on fixed subscriptions.

The expanded data ecosystem provides a foundation for using Rwanda's official statistics, education information, workforce intelligence, opportunity announcements and other permitted sources.

The project should evolve through a disciplined implementation process: a working hackathon MVP, a validated national pilot, institutional adoption and eventually regional or global expansion.

HDOS Rwanda™ is not intended to be merely a career website, a learning platform or an AI chatbot.

It is designed as a Human Development Intelligence Operating System that connects goals, evidence, decisions, opportunities, applications and measurable progress.

Its central purpose is to help people transform potential into practical possibilities and possibilities into meaningful development.

HDOS RWANDA™

From Potential to Possibility. From Possibility to Progress.

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32. APPENDICES

APPENDIX A — PRIORITY OFFICIAL DATA SOURCES

The following sources should form the starting point of the HDOS Rwanda data registry.

Source	Intended use	Official reference
NISR Labour Force Survey	Employment and workforce indicators	https://statistics.gov.rw/data-sources/surveys/Labour-Force-Survey
NISR Labour Force Survey 2026	Current quarterly labour information	https://statistics.gov.rw/data-sources/surveys/Labour-Force-Survey/labour-force-survey-2026
NISR Statistical Yearbook 2025	National and sector statistics	https://www.statistics.gov.rw/statistical-publications/statistical-yearbook/rwanda-statistical-yearbook-2025
NISR EICV7	Poverty and household indicators	https://gis.statistics.gov.rw/data-sources/surveys/EICV/integrated-household-living-conditions-survey-7-eicv-7

MINEDUC Education Statistics	Education and TVET statistics	https://www.mineduc.gov.rw/
Higher Education Council	Accredited institutions and scholarships	https://www.hec.gov.rw/
Rwanda LMIS	Occupations, skills and workforce information	https://lmis.rw/
Kora	Job access and job matching	https://www.kora.mifotra.gov.rw/
Internship Rwanda	Internship information	https://www.internship.rw/
Rwanda Development Board	Sector and investment opportunities	https://rdb.rw/investment-opportunities/
Rwanda Data Protection and Privacy Office	Data-protection requirements	https://dpo.gov.rw/

Every source must be checked for its terms of access, reuse permissions, licensing and update schedule before integration.

APPENDIX B — OFFICIAL NATIONAL DEVELOPMENT REFERENCES

- Second National Strategy for Transformation (NST2), 2024–2029.
- Rwanda Vision 2050.
- National education and skills-development strategies.
- Relevant employment and private-sector development priorities.
- Applicable national digital-transformation policies.

APPENDIX C — LEGAL AND PRIVACY REFERENCE

Law No. 058/2021 of 13 October 2021 relating to the protection of personal data and privacy.

Official information and guidance:

https://dpo.gov.rw/

The project should obtain appropriate legal and data-protection review before launching personal-data processing at scale.

APPENDIX D — SAMPLE USER DEVELOPMENT PATHWAY

Goal: Become an AI Engineer.

Stage	Development action	Expected evidence
Assessment	Review education, mathematics and programming knowledge	Skills profile
Foundation	Learn Python and programming fundamentals	Practical exercises
Technical development	Learn statistics, data handling and machine learning	Completed learning activities
Practical development	Build AI projects	Project portfolio
Professional readiness	Improve CV, communication and collaboration	Professional profile
Opportunity access	Search relevant internships and entry-level positions	Opportunity shortlist
Application	Prepare and submit selected applications	Application records
Continuous development	Monitor new skills and labour-market requirements	Updated roadmap

The pathway is illustrative and must be personalized.

APPENDIX E — SAMPLE OPPORTUNITY RECORD


```
{
  "opportunity_id": "OPP-EXAMPLE-001",
  "title": "Example Scholarship",
  "provider": "Official Publishing Organization",
  "category": "SCHOLARSHIP",
  "country": "Rwanda",
  "education_level": "UNDERGRADUATE",
  "description": "Illustrative record for system development.",
  "eligibility_rules": [],
  "required_documents": [],
  "application_url": null,
  "source_url": null,
  "published_at": null,
  "deadline": null,
  "last_verified_at": null,
  "verification_status": "SAMPLE_DATA",
  "status": "DRAFT",
  "is_demo": true
}
```

APPENDIX F — SAMPLE SERVICE-CONSUMPTION RECORD


```
{
  "service_usage_id": "USE-EXAMPLE-001",
  "user_id": "USER-EXAMPLE-001",
  "service_code": "SCHOLARSHIP_ELIGIBILITY_ASSESSMENT",
  "service_version": "1.0",
  "price_amount": 0,
  "currency": "RWF",
  "credit_cost": 4,
  "status": "DEMO",
  "reserved_at": null,
  "completed_at": null,
  "delivered_artifact_id": null,
  "refund_status": "NOT_APPLICABLE",
  "is_demo": true
}
```


APPENDIX G — GLOSSARY

AI: Artificial Intelligence.

API: Application Programming Interface.

EICV: Integrated Household Living Conditions Survey.

HEC: Higher Education Council.

HDOS: Human Development Intelligence Operating System.

LMIS: Labour Market Information System.

Life DNA: HDOS personal development profile.

MVP: Minimum Viable Product.

NISR: National Institute of Statistics of Rwanda.

NST2: Second National Strategy for Transformation, 2024–2029.

Pathway Fit Index: HDOS proposed explainable assessment of the alignment between a user's circumstances and a selected pathway.

TVET: Technical and Vocational Education and Training.

Usage Metering: Recording the consumption of billable services.

Data Provenance: Information describing where data originated, its reference period, publication and verification history.

FINAL EDITORIAL AND IMPLEMENTATION PRINCIPLE

HDOS should always distinguish:

1. What the original proposal defines.
2. What has been added in this revised design.
3. What is actually implemented and tested.
4. What is still planned or requires validation.

The quality of HDOS will depend not on the number of features displayed, but on the accuracy of its data, the usefulness of its recommendations, the transparency of its decisions, the security of users' information and the real progress it helps users make.

END OF CONSOLIDATED HDOS RWANDA PROJECT BOOK — VERSION 2.0